

BYN2 – BYN2 TASK 1: DEPLOYING A MODIFIED MULTITHREADED SPRING APPLICATION TO THE CLOUD

ADVANCED JAVA – D387

PRFA – BYN2

Preparation

Task Overview

Submissions

Evaluation Report

COMPETENCIES

4104.1.1 : Writes Multithreaded Code

The learner writes multithreaded, object-oriented code using Java frameworks.

4104.1.2 : Deploying Applications with Cloud Services

The learner determines how to deploy software applications using cloud services.

INTRODUCTION

Throughout your software design and development career, you will be asked to create, customize, and maintain applications with various features and functionality based on business requirements. For this assessment, you will modify a Spring application with a Java back end and an Angular front end to include multithreaded language translation, a message at different time zones, and currency exchange. Then, you will build a Docker image of the current multithreaded Spring application and containerize it using the supporting document and web links provided.

The skills you showcase in your completed application will be helpful in responding to technical interview questions for future employment. This application may also be added to your portfolio to show to prospective employers.

Your submission should include a link to the project that contains the Dockerfile, a copy of the repository branch history file, and the URL to the GitLab repository for evaluation. The submission must keep the project file and folder structure intact for the integrated development environment (IDE).

Note: The IDE for this assessment is IntelliJ IDEA (Ultimate Edition). Instructions on how to download this is provided in the attached supporting document.

SCENARIO

You are working for a company located in Toronto, Canada, that schedules hotel reservations. As a software developer, your job is to modify the Landon Hotel scheduling application to meet new requirements under new management. You will choose any application user you would like.

You have been provided a Spring application with a Java back end, an Angular front end, and web links and a supporting document to assist you in your work.

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker. Students are strongly encouraged to wait for the similarity report to generate after uploading their work and then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See [Understanding Similarity Reports](#) for more information.

Grammarly Note:

Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged to review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See [Use Grammarly for Education Effectively](#) for more information.

Microsoft Files Note:

Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see [Computer System and Technology Requirements](#).

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

Note: External plug-ins and libraries other than those specified in this assessment are not allowed.

A. Create your subgroup and project in GitLab using the provided web link and the "GitLab How-To" web link by doing the following:

- Clone the project to the IDE.
- Commit with a message and push when you complete *each* requirement listed in parts B1, B2, B3, and C1.

Note: You may commit and push whenever you want to back up your changes, even if a requirement is not yet complete.

- Submit a copy of the GitLab repository URL in the "Comments to Evaluator" section when you submit this assessment.
- Submit a copy of the repository branch history retrieved from your repository, which must include the commit messages and dates.

Note: Wait until you have completed all the following prompts before you create your copy of the

repository branch history.

B. Modify the Landon Hotel scheduling application for localization and internationalization by doing the following:

1. Install the Landon Hotel scheduling application in your integrated development environment (IDE). Modify the Java classes of application to display a welcome message by doing the following:
 - a. Build resource bundles for *both* English and French (languages required by Canadian law). Include a welcome message in the language resource bundles.
 - b. Display the welcome message in *both* English and French by applying the resource bundles using a different thread for each language.

Note: You may use Google Translate for the wording of your welcome message.

2. Modify the front end to display the price for a reservation in currency rates for U.S. dollars (\$), Canadian dollars (C\$), and euros (€) on different lines.

Note: It is not necessary to convert the values of the prices.

3. Display the time for an online live presentation held at the Landon Hotel by doing the following:
 - a. Write a Java method to convert times between eastern time (ET), mountain time (MT), and coordinated universal time (UTC) zones.
 - b. Use the time zone conversion method from part B3a to display a message stating the time in *all* three times zones in hours and minutes for an online, live presentation held at the Landon Hotel. The times should be displayed as ET, MT, and UTC.

C. Explain how you would deploy the Spring application with a Java back end and an Angular front end to cloud services and create a Dockerfile using the attached supporting document "How to Create a Docker Account" by doing the following:

1. Build the Dockerfile to create a single image that includes *all* code, including modifications made in parts B1 to B3. Commit and push the final Dockerfile to GitLab.
2. Test the Dockerfile by doing the following:
 - Create a Docker image of the current multithreaded Spring application.
 - Run the Docker image in a container and give the container a name that includes D387_[student ID].
 - Submit a screenshot capture of the running application with evidence it is running in the container.
3. Describe how you would deploy the current multithreaded Spring application to the cloud. Include the name of the cloud service provider you would use.

Note: Remember to commit and push your changes to GitLab.

D. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ()

File size limit: 200 MB

File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A:GITLAB REPOSITORY

NOT EVIDENT

A GitLab repository is not provided.

APPROACHING COMPETENCE

The subgroup and project are created in GitLab, but 1 or more of the given actions are not completed, or they are completed incorrectly.

COMPETENT

The subgroup and project are created in GitLab correctly, and *all* of the given actions are completed correctly.

B1A:LANGUAGE RESOURCE BUNDLES

NOT EVIDENT

The Landon Hotel scheduling application does not include language resource bundles, or no code is provided.

APPROACHING COMPETENCE

The Landon Hotel scheduling application includes language resource bundles, but the bundles are not for *both* English and French. A welcome message is not built into the language resource bundles. Or the code contains errors or is incomplete.

COMPETENT

The Landon Hotel scheduling application includes resource bundles for *both* English and French, including a welcome message built into the language resource bundles. The code is complete and functions correctly.

B1B:DISPLAY WELCOME MESSAGE

NOT EVIDENT

The Landon Hotel scheduling application does not display a welcome message, or no code is provided.

APPROACHING COMPETENCE

The Landon Hotel scheduling application has limited functionality to display a welcome message, or the language resource bundles do not use a different thread to display the languages in *both* English and French. Or the application does not use the Java classes resource bundles to display the languages. Or the code contains errors or is incomplete.

COMPETENT

The Landon Hotel scheduling application has the functionality to display a welcome message in *both* English and French by applying the resource bundles in the Java classes using a different thread for *each* language. The code is complete and functions correctly.

B2:DISPLAY CURRENCY RATES

NOT EVIDENT

The Landon Hotel scheduling application does not display a price for a reservation, or no code is provided.

APPROACHING COMPETENCE

The Landon Hotel scheduling application displays the price for a reservation, but 1 or more of the required currency rates are missing or are not on different lines. Or the code contains errors or is incomplete.

COMPETENT

The Landon Hotel scheduling application has the functionality to display the price for a reservation in currency rates for U.S. dollars (\$), Canadian dollars (C\$), and euros (€) on different lines. The code is complete and functions correctly.

B3A:TIME ZONE CONVERSION METHOD**NOT EVIDENT**

A time zone conversion method is not provided.

APPROACHING COMPETENCE

The Java method incorrectly converts the time between eastern time, mountain time, and coordinated universal time zones. Or 1 or more of the required time zones is missing. Or the code contains errors or is incomplete.

COMPETENT

The Java method correctly converts the time between eastern time, mountain time, and coordinated universal time zones. The code is complete and functions correctly.

B3B:DISPLAY TIME ZONES**NOT EVIDENT**

The Landon Hotel scheduling application does not display the time for an online, live presentation held at the hotel, or no code is provided.

APPROACHING COMPETENCE

The Landon Hotel scheduling application displays the time of an online, live presentation held at the hotel, but 1 or more of the required time zones from part B3a are missing. Or the time zones are not correctly displayed as ET, MT, and UTC. Or the code contains errors or is incomplete.

COMPETENT

The Landon Hotel scheduling application displays the time of an online, live presentation held at the hotel in *all* 3 time zones from part B3a in hours and minutes. The times are correctly displayed as ET, MT, and UTC. The code is complete and functions correctly.

C1:BUILD DOCKERFILE**NOT EVIDENT**

A Dockerfile is not present.

APPROACHING COMPETENCE**COMPETENT**

The Dockerfile image is successfully built from the Dockerfile

The Dockerfile image is present but does not successfully build from the Dockerfile, or it does not include *all* code and modifications made in parts B1 to B3.

and includes *all* code and modifications made in parts B1 to B3.

C2:TEST THE DOCKERFILE

NOT EVIDENT

A Dockerfile is not tested.

APPROACHING COMPETENCE

The Dockerfile is tested but does not include *all* the given requirements, or some of the listed requirements contain errors.

COMPETENT

The Dockerfile is tested and includes *all* given requirements without errors.

C3:DESCRIPTION OF DEPLOYMENT TO THE CLOUD

NOT EVIDENT

The description does not explain how the current multithreaded Spring application could be deployed. Or the description of deployment to the cloud is not provided.

APPROACHING COMPETENCE

The description explains how the current multithreaded Spring application could be deployed but does not plausibly align with the given cloud provider services. Or a cloud service provider is not included.

COMPETENT

The description provides a plausible explanation of how the current multithreaded Spring application could be deployed to the cloud that aligns with the given cloud provider services.

D:PROFESSIONAL COMMUNICATION

NOT EVIDENT

This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, negatively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.

APPROACHING COMPETENCE

This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.

COMPETENT

This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote accurate interpretation and understanding.

WEB LINKS

[Docker](#)

[GitLab](#)

Access Landon Hotel Scheduling Application—Integrated Spring Back End and Angular Front End for IntelliJ IDEA in this environment

[GitLab How-To](#)

[IntelliJ IDEA \(Ultimate Edition\) IDE – Student License](#)

[Node.js](#)

[How to Create a Docker Account](#)

SUPPORTING DOCUMENTS

[IntelliJ Ultimate Edition Directions.docx](#)